

AnyCam 3.7.0-rc1.0 Audit Report

Release audit - 04 October 2026 - hardware decode diagnostics - tag 3.7.0-rc1.0

Scope: phase 3.7.0 of the build plan: B11 and C9, diagnostics only. CrystalHeeler, 2026-10-04: build the diagnostics; the fix needs the logs (CLAUDE.md rule 2). The last phase of the untested run, on top of 3.6.0-rc1.0. This build is not field-tested.

1. What the diagnostics show

Part	Design
Start-up report	Every <code>/dev/video*</code> , <code>/dev/media*</code> and <code>/dev/dri/renderD*</code> device inside the add-on: its V4L2 name from <code>/sys/class/video4linux</code> , and whether it opens for reading and writing (else the system's reason, such as "Operation not permitted"). Logged even with Hardware Decode off.
C9	No device named <code>rpivid</code> : a warning names the line <code>dtoverlay=rpivid-v4l2</code> for <code>/boot/firmware/config.txt</code> . A toggle is not possible from inside the add-on: it cannot write the boot partition.
B11	A device that does not open is named, with the note that hardware decode on it falls back to software.
The first-frame line	ffmpeg's error lines are read while it runs. A line that says a hardware decoder did not open (a device or hwaccel subject with a failure word) is kept for that camera's current start. The first picture is then logged as "decoded in SOFTWARE" with ffmpeg's line, not as "hw first frame".
<code>/api/diagnostics/hw</code>	The report, the Hardware Decode setting, the decoder candidates, those found unavailable, and each camera's software fallback.

Why this first: the field log behind B11 showed ffmpeg failing to open `/dev/media0` to `/dev/media3` with "Operation not permitted" while AnyCam logged hardware decode. The cause can be the add-on's device access, a Home Assistant OS change, or the ffmpeg version; the report separates these in one start-up log.

Tests run

Suite	Result	Scope
Release gate	PASS	Eight gates; 17 files; build inputs checked online.
Python integration	PASS	568 of 568; 8 new: AH1 (the real snapshot loop with a fake ffmpeg that falls back), AH2 to AH5.
JavaScript behaviour	PASS	158 of 158, unchanged.
Undefined names	PASS	17 files, 0.
Not run	NOTED	No field test (CrystalHeeler's order). The image was not built. The device report ran against stand-in devices, not a Pi; the ffmpeg failure lines are from the B11 field log and ffmpeg's source wording, not from this build.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*, Parts 1 to 6. Rows this change touches; the rest carry from the 3.6.0-rc1.0 audit.

Standard	Check applied	Result	Note
1.1 Truncated function	AST, contracts, undefined names	PASS	17 files.
1.4 Async	Blocking I/O	PASS	The device report runs in a thread.
1.12 Logging	Truth	PASS	The log no longer claims hardware decode that did not run.
1.18 Testing	New tests	PASS	8 server checks.
6.1 Security	Devices	PASS	Each device is opened and closed at once, without reading or writing; the ffmpeg line is cleaned of passwords.
Rule 9 (privacy)	New text	PASS	No addresses or names.

The untested run, complete

Release candidate	Items	Server / page checks
3.0.1-rc1.0	B2, C10, B20, B23, B24, B26, B12, B8, F10, C11 (B3), C20	471 / 118
3.1.0-rc1.0	B25, C19, D3	494 / 132
3.2.0-rc1.0	C5, C6	503 / 139
3.3.0-rc1.0	C4, B15	513 / 139
3.4.0-rc1.0	C17	532 / 158
3.5.0-rc1.0	D1, D2	542 / 158
3.6.0-rc1.0	C14	560 / 158
3.7.0-rc1.0	B11, C9 (diagnostics)	568 / 158

Each is its own commit, tag and zip; none is pushed or published.

Known issues carried forward

- **B11:** hardware decode can still fall back to software; the fix waits for the 3.7.0-rc1.0 start-up log.
- **B6:** the Microseven waits until CrystalHeeler unlocks it.
- **C21:** the classic engine stays as the fallback.

Acceptance (field test)

- The start-up log has one "HW device" line for each decoder device, and says whether each opens.
- Enhanced View on the 4K DVR channel in the classic view: the log says either "hw first frame" or "decoded in SOFTWARE" with a reason.

- /api/diagnostics/hw answers in the browser through Home Assistant.

Files changed

File	Change
camera_discovery.py	_hw_device_report, _hw_report_log, api_diagnostics_hw; the probe logs the report; version 3.7.0-rc1.0.
anycam_snap.py	_hw_fallback_line; the stderr reader keeps the line; the first-frame log.
tests/	AH1 in the snapshot loop section; section AH.
CHANGELOG.md, config.yaml, docs	3.7.0-rc1.0; build plan.